

Sales Forecasting with Time Series Analysis — Project Report

1. Time Series Analysis

Data. 730 days (2 years) of daily `sales_units`, synthetically generated with an upward trend, a 7-day weekly seasonal cycle, a secondary 365-day yearly cycle, and Gaussian noise — designed to resemble realistic retail sales behavior rather than a random walk.

Decomposition. Additive seasonal decomposition (7-day period) confirmed all three expected components: a steady upward trend across the two years, a clear repeating weekly seasonal pattern, and residual noise with no obvious remaining structure.

Stationarity. An Augmented Dickey-Fuller test on the raw series returned an ADF statistic of -0.481 (p-value 0.896), confirming the series is **non-stationary** — expected, given the built-in trend and seasonality. This directly motivated using SARIMA (which differences internally, order $(1, 1, 1)$, seasonal order $(1, 1, 1, 7)$) and Exponential Smoothing (which models trend and seasonality explicitly) rather than a naive model that assumes stationarity.

2. Modeling Approach

Four forecasting approaches were implemented and compared on a strict time-based 80/20-style split (last 28 days held out as test, no shuffling — critical for valid time series evaluation):

- **Moving Average** (7-day window, walk-forward) — simple baseline.
- **Exponential Smoothing** (Holt-Winters, additive trend + additive weekly seasonality).
- **SARIMA** — order $(1,1,1)$, seasonal order $(1,1,1,7)$, providing native 95% confidence intervals.
- **Random Forest on lagged features** (7 lag features, ML-based alternative approach).

3. Results

Model	MAE	RMSE	MAPE
Moving Average	10.157	11.496	8.79%
Exponential Smoothing	**4.253**	**5.101**	**3.69%**
SARIMA	4.933	6.322	4.26%
Random Forest (lagged)	6.861	8.306	5.84%

Best model: Exponential Smoothing, with the lowest error across all three metrics (MAE 4.25, RMSE 5.10, MAPE 3.69%). This is a sensible outcome — Holt-Winters is purpose-built for series with a clear additive trend and fixed seasonal period, which matches exactly how this dataset was

constructed. SARIMA performed close behind (MAPE 4.26%), which is expected since it explicitly handles the same trend/seasonality structure via differencing and seasonal terms. The Moving Average baseline, as expected, performed clearly worst (MAPE 8.79%) since it has no mechanism to capture trend or seasonality — its role here is purely as a sanity-check floor. The Random Forest on lagged features sat between the classical models and the baseline (MAPE 5.84%): a reasonable result, but tree-based models with a small, fixed lag window are inherently less suited to periodic structure than Holt-Winters/SARIMA, which model seasonality directly.

Unlike Tasks 1-4, none of these results are artificially perfect — errors are realistic and differentiated across models, which strengthens confidence that this evaluation reflects genuine forecasting skill rather than a synthetic-data artifact.

4. Forecast — Next 4 Weeks (with confidence intervals)

Using SARIMA fit on the full 730-day history, the model forecasts daily sales for the next 28 days (Dec 31, 2024 – Jan 27, 2025), with the weekly seasonal cycle clearly visible in the predictions (values oscillate roughly between ~103 and ~132 units, repeating on a 7-day rhythm consistent with the historical pattern):

Date	Forecast (units)
2024-12-31	130.4
2025-01-01	123.3
2025-01-02	111.3
2025-01-03	102.5
2025-01-04	104.7
2025-01-05	117.6
2025-01-06	128.9
...	*(repeats weekly pattern through 2025-01-27)*
2025-01-27	130.9

95% confidence intervals for each forecasted day are computed by the SARIMA model (`get_forecast().conf_int()`) and plotted alongside the point forecast in `report/forecast_vs_actual.png`, widening appropriately further into the forecast horizon — reflecting increasing uncertainty the further out the prediction extends.

5. Challenges

- **Non-stationarity** required explicit handling (SARIMA differencing, Exponential Smoothing's trend/seasonal decomposition) rather than naive modeling — confirmed via the ADF test before

model selection.

- **Model selection trade-off:** while SARIMA is the more theoretically rigorous choice (native confidence intervals, explicit statistical structure), Exponential Smoothing outperformed it slightly on this dataset — a reminder that simpler models can beat more complex ones when their assumptions match the data well.
- **Lag-based ML approach's ceiling:** Random Forest with only lag features lacks an explicit seasonal mechanism, which likely explains its weaker performance relative to the classical time series models on this clearly seasonal dataset.

6. Potential Improvements

- Extend evaluation with time series cross-validation (rolling-origin, e.g. `sklearn.model_selection.TimeSeriesSplit`) rather than a single train/test split, for a more robust performance estimate.
- Add calendar-based and holiday features to the Random Forest approach (day-of-week, month, holiday flags) to give it comparable seasonal signal to SARIMA/Exponential Smoothing.
- Try Prophet or a gradient-boosted time series model (e.g. LightGBM with engineered seasonal features) as an additional comparison point.
- Validate against real historical sales data, since the synthetic data's seasonality is clean and consistent — real sales are typically noisier and may include irregular effects (promotions, holidays, stockouts) not present here.